

SQL Analysis Documentation

SQL patterns, analytical views, and validation checks

1. SQL Objectives

- Validate record counts and classifications.
- Summarize by priority and category.
- Analyze urgency and risk indicators.
- Provide reusable datasets for Power BI.

2. Example Queries

```
SELECT priority_name, COUNT(*) AS ticket_count
FROM vw_ticket_summary
GROUP BY priority_name;

SELECT topic_group, COUNT(*) AS ticket_count
FROM vw_ticket_summary
GROUP BY topic_group
ORDER BY ticket_count DESC;

SELECT COUNT(*) AS critical_tickets
FROM vw_ticket_summary
WHERE priority_name = 'Critical';

SELECT COUNT(*) AS security_tickets
FROM vw_ticket_summary
WHERE security_issue = TRUE;
```

3. Analytical Views

View	Purpose
vw_ticket_summary	Primary Power BI reporting dataset.
vw_category_kpis	Category-level KPI aggregation.
vw_priority_summary	Priority-level summary.
vw_high_impact_tickets	High-impact/investigation-oriented subset.

4. Validation

- Critical count validated at 321.
- High-urgency threshold validated at urgency_score >= 70; count = 321.
- Urgency distribution by priority cross-checked against Power BI.
- Critical-ticket SQL results cross-checked against Power BI analysis.